

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Seventh Semester of B. Tech. (CL) Examination
May - 2013

CL 401 Estimation, Costing & Financial Concepts - I

Date: 07.05.2013, Tuesday

Time: 10:00 p.m. To 1:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Calculate following major quantities for Shop construction. Details are shown in Figure 1. **12**

- i. Excavation work
- ii. PCC work with M10 Concrete.
- iii. Brick work in foundation(CM 1:6) and super structure(CM 1:4)
- iv. D.P.C. work (1:2:4) 2 coat bitumen laid hot.
- v. 10 mm flooring tiles over 10 cm cement concrete and 10 cm sand filling

Q - 2 Attempt any two

12

- (a) Define rate analysis and explain various factors affecting it.
- (b) Workout the rate analysis of R.C.C. beam proportion (1:2:4) with steel reinforcement for 1cu m. Assume suitable data if required.
- (c) Workout the rate analysis of 1st class brickwork in CM (1:6) up to plinth using modular bricks – for 10 cu m.

Q - 3 Attempt any one

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- (a) Calculate following quantities for chajja shown in Figure 2. Consider clear span of 1.2 m, breadth 0.6 m, average thickness of 10 cm, resting on 150 mm wall. Assume suitable data if required.
 - i. RCC work
 - ii. Steel quantity
 - iii. Plastering
 - iv. Pointing
- (b) Calculate following quantities for compound wall shown in Figure 3(a) and 3(b). Consider height of 1.2 m from G.L. Assume suitable data if required.
 - i. Excavation work
 - ii. PCC work
 - iii. Brick masonry work
 - iv. Plastering work

SECTION – II

- Q - 4 (a) State the general principle for deciding the unit of measurement for various items of work. 03
- (b) List the detail require for preparation of estimation. 04
- Q - 5 (a) Differentiate between brief specifications and detailed specifications. 05
- (b) Write detailed specifications for Cement concrete (1: 4: 8) work in foundation. 05
- (c) State the general mode of measurement and payment for excavation work. 04
- OR**
- Q - 5 (a) Define specifications. Give the principles of writing specifications. 05
- (b) Write detailed specifications for Damp proof course (D.P.C.) work. 05
- (c) State the kind and mode of deduction form measurement of Pointing and Plastering. 04
- Q - 6 (a) State the information to be given in a tender notice. 05
- (b) Explain what is meant by contract and give details of various forms used for agreements. 05
- (c) Distinguish clearly between revised estimate and supplementary estimate 04
- OR**
- Q - 6 (a) Draft conditions of contract for Work at night and holidays 05
- (b) What is open tender? Also state advantages and drawbacks of same. 05
- (c) Define technical terms: Competent Authority, Contingencies charges, Lumpsum items and Over head cost. 04

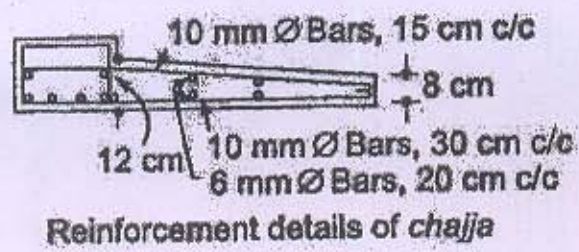


Figure 2. Section of Chajja.

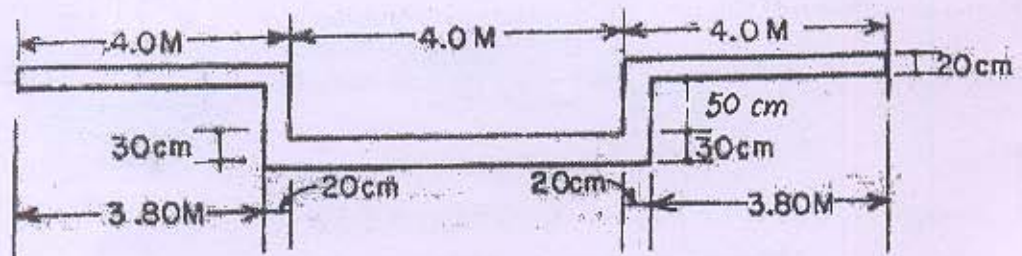


Figure 3(a). Plan of a compound wall

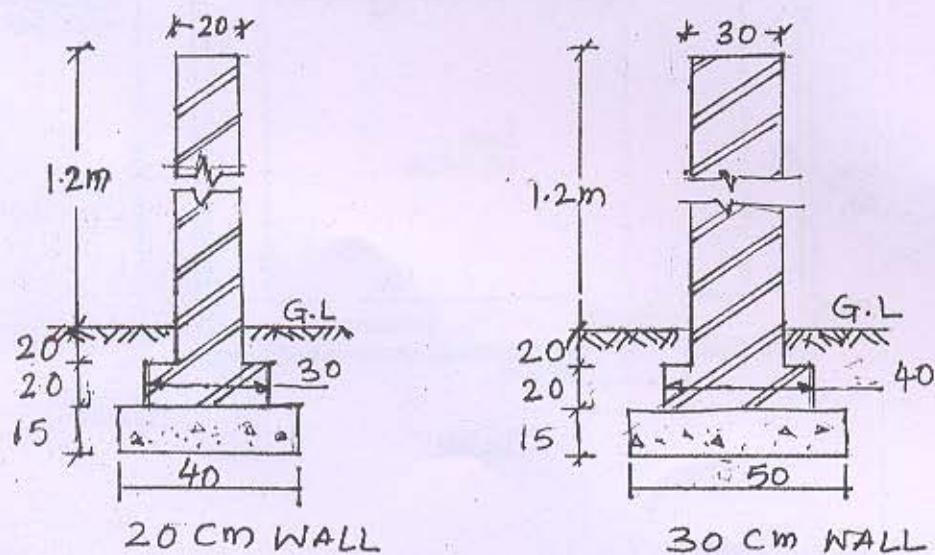


Figure 3(b). Section of a compound walls
